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Systematic Review and Meta-analysis

Pulmonary Hypertension Association with mortality in Newborns with Giant Omphalocele: A Systematic Review and Meta-Analysis

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Abstract:

Introduction: Pulmonary hypertension (PH) is associated with adverse outcomes in infants with omphalocele, but existing studies are limited by small sample sizes and single-center designs. This meta-analysis evaluated the association between PH and mortality, as well as secondary clinical outcomes.

Methods: A systematic search in MEDLINE, EMBASE, and Cochrane CENTRAL identified studies of patients with giant omphalocele stratified by PH status. Eligible studies included nonrandomized cohorts, cross-sectional studies, or case controls assessing mortality in this setting. Exclusions were overlapping populations or incomplete data. Primary outcome was mortality. Secondary outcomes included duration of mechanical ventilation (MV), length of hospital and Neonatal Intensive Care Unit (NICU) stay, parenteral nutrition, tracheostomy, high-frequency ventilation, and home oxygen. Risk of bias was assessed using the Newcastle-Ottawa Scale (NOS). The study protocol was registered in PROSPERO (CRD420251025073), and there was no funding for this study.

Results: Of 238 studies screened, five retrospective cohorts met criteria, totaling of 258 patients (112 [43%] with PH and 146 [57%] without). PH was associated with increased mortality (OR 6.78, 95% CI 2.72–16.88, $p < 0.001$, $I^2 = 0\%$) and prolonged MV (mean difference 21.25 days, 95% CI 13.19–29.31, $p < 0.001$, $I^2 = 0\%$). Limitations included publication bias, incomplete reporting of clinical variables and the lack of standardized criteria.

Conclusion: PH in neonates with omphalocele is associated with higher mortality and prolonged MV. Future studies should clarify risk factors and develop targeted management strategies to improve survival and long-term outcomes.

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Introduction

Omphalocele is a congenital abdominal wall defect characterized by herniation of abdominal viscera into the base of the umbilical cord[1-6]. Although relatively rare and with an incidence estimated between 1 in 2,500 to 1 in 6,000 live births[4-9], the diagnosis carries high morbidity and mortality, particularly when identified in the first trimester, where reported mortality rates range from 30% to 52%[3,7]. Prognosis depends on multiple factors, including defect size, associated structural or chromosomal anomalies, and complications such as pulmonary hypoplasia [1,4,9-12].

Pulmonary hypertension (PH) is increasingly recognized as a significant comorbidity in neonates with omphalocele, particularly in cases of giant defects requiring staged repair and prolonged ventilatory support[3-6,13-17]. Etiology is likely multifactorial, involving reduced thoracic volume from herniated viscera, systemic inflammation, and abnormal cardiopulmonary hemodynamics[1,4,6,10-12]. PH in this setting has been associated with prolonged hospitalization, greater need for advanced cardiorespiratory support such as extracorporeal membrane oxygenation (ECMO), and increased mortality risk[3,5,13,16]. Despite these observations, existing studies are limited by small sample sizes, and they are mostly single-center studies, restricting the ability to draw more generalizable conclusions.

To address this gap, we conducted a systematic review and meta-analysis comparing outcomes in patients with giant omphalocele, stratified by the presence or absence of PH. The primary aim is to evaluate the association between PH and mortality. Secondary aims included identifying additional risk factors for poor outcomes, with the goal of improving prenatal counseling, optimizing delivery and postnatal management strategies, and informing future research in this high-risk neonatal population.

Methods

A protocol was established and prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO), number CRD420251025073[18]. This review was conducted in line with the recommendations from the Cochrane Collaboration (Cochrane Guidelines)[19] and the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guidelines[20]. The PRISMA checklist is provided as Supplementary Material.

Search Strategy

We systematically searched PubMed (MEDLINE), Embase, Cochrane Database of Systematic Reviews and Latin American and Caribbean Health Sciences literature (LILACS). We used the keywords “omphalocele”, “exomphalos”, “abdominal wall defect”, “pulmonary hypertension”, “pulmonary arterial hypertension”, “pulmonary vascular disease”. The search strategy is available in the Supplementary Material. We also searched for the references within the articles included in the study.

Study Selection and Eligibility

Studies were included if they met the following criteria: (i) nonrandomized cohorts, cross-sectional studies, or case controls; (ii) assessing the relation between PH and mortality; (iii) in patients with PH and omphalocele; and (iv) reporting at least one of the outcomes of interest. We excluded studies with overlapping populations, considered derived from overlapping institutions and recruitment periods, in which case only the study with the largest sample was selected for inclusion in the meta-analysis. Case reports and case series were excluded. Abstracts were only included if they provided sufficient information for data collection. There was no language restriction. When the data required was not available in the manuscript, we emailed the corresponding authors.

Data Collection

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Two reviewers (L.M.I.M. and S.G.) independently performed study selection. In cases of disagreement of the two reviewers, we involved a third author (G.A.D.C.). Data extraction was performed by two independent authors (L.M.I.M. and S.G.) and was reviewed by a third author (G.A.D.C.). Data extracted included: study design, location, period of the study, general demographic information, inclusion and exclusion criteria, diagnostic criteria, mortality, days of mechanical ventilation (MV), length of hospital stay, birth at weight, gestational weight, need for inhaled nitric oxide, need for parenteral nutrition, presence of other cardiac and congenital abnormalities and sex.

Definitions

All studies included in the meta-analysis focused exclusively on giant omphaloceles. The definitions of pulmonary hypertension used in each manuscript is summarized on Table 1.

Outcomes

The primary outcome of interest was the association between the presence of PH in patients with giant omphalocele and increased mortality.

Secondary outcomes included: length of MV, length of hospital and Neonatal Intensive Care Unit (NICU) stay, need for parenteral nutrition, need for tracheostomy, need for high-frequency ventilation, and need for home oxygen support. We also looked for demographic differences between the two groups including: gestational age, weight at birth, sex, Apgar, presence of other congenital abnormalities.

Quality Assessment

Quality assessment of included studies was performed by two independent authors (G.A.D.C. and L.R.C.) using the Newcastle-Ottawa Scale (NOS)[21]. The Newcastle-Ottawa Scale (NOS) is used for assessing the risk of bias in observational studies. Any discrepancy was resolved by involvement of a third author (L.M.I.M.). The NOS evaluates three domains: Selection (four items), Comparability (one item), and Outcomes (three items). Each item was awarded a score if the criterion was met, with a maximum of nine. As recommended, studies were considered as high-quality if their NOS score was six or more; as moderate-quality if three to five; and low-quality if two or less.

Statistical Analysis

For continuous outcomes, we performed the generic inverse variance method to compare weighted mean differences as recommended per Cochrane[19]. Dichotomous outcomes were analyzed using odds ratio (OR) with 95% CI also using the generic inverse variance method. Whenever available, adjusted OR and their respective model were used. When not available, the crude OR was used. Since high clinical, methodological and statistical heterogeneity was expected due to the retrospective nature of the studies, we used random-effect models. Statistical heterogeneity was examined with Cochran's Q test and I^2 statistics. In cases where significant heterogeneity was considered ($I^2 > 50\%$), we planned to perform a leave-one-out sensitivity analysis as well as Baujat evaluation. All statistical analysis were in accordance with Cochrane Collaboration (Cochrane Guidelines)[19]. Statistical analysis was performed using Review Manager (RevMan) software, version 5.4.

Publication Bias

For analysis with more than ten studies, we planned to perform inspection for publication bias using Begg's funnel plot and Egger test. P-values inferior to 0.05 were considered indicative of publication bias.

GRADE Assessment

The Grading of Recommendations, Assessment, Development, and Evaluation (GRADE)[22] tool was used by three independent authors (G.A.D.C., L.M.I.M. and E.M.H.P.) to assess the level of certainty of the evidence in this review, with categorizations ranging from high to very low (Supplementary Material – Table S1). Disagreements were resolved by consensus.

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Results

Search Strategy and Study Selection:

We identified 238 studies and obtained 189 studies after removing duplicates. A total of 25 studies met our initial eligibility criteria, which were analyzed in full. The most common exclusion criteria were presence of hemodynamically significant congenital heart diseases and other abdominal wall abnormalities. After careful revision, four studies were included. One additional study was added, after de-identified additional data was obtained from one of the authors reached out, leading to a total of 5 studies in our meta-analysis. The PRISMA flow diagram is available in Figure 1.

Study and Patients Baseline Characteristics:

All the five studies were retrospective in nature including patients from the United States, China, and France[4-6, 8, 16]. We included 258 neonates with giant omphalocele across the five studies. PH was present in 112 (43.4%) and 146 (56.6%) did not have PH. The summary of the included studies and the characteristics of the study population are summarized, respectively, in Tables 1 and 2.

Primary Outcomes:

The pooled analysis of our primary outcome found an increased odds ratio (OR) of 6.78 (95% CI 2.72 -16.88, $p < 0.001$, $I^2 = 0\%$) for mortality in patients with PH when compared to patients with no PH (shown in Figure 2).

Secondary Outcomes:

The pooled analysis for duration of MV revealed a mean difference of 21.25 extra days (95% CI 13.19 - 29.31, $p < 0.001$, $I^2 = 0\%$) in patients with PH when compared to patients with no PH (shown in Figure 3). Unfortunately, we were not able to meta-analyze pre-established outcomes such as Apgar at birth, length of hospital stay, length of NICU stay, need for gastrostomy, parenteral nutrition, tracheostomy, high-frequency ventilation and home oxygen support due to the lack of data. There was no statistically significant difference between groups regarding median gestational age (MD -0.19; 95% CI -0.88 - 0.50, $p = 0.59$, $I^2 = 14\%$). The forest plot is available on Figure S1. There was no association between male sex and PH with an OR of 1.27 (95% CI 0.66 – 2.43, $p = 0.48$, $I^2 = 33\%$), available on Figure S2. We did not find a statistically significant mean difference in weight a birth in patients with or without PH (MD 0.01, 95% CI -0.18 - 0.20, $p = 0.92$, $I^2 = 0\%$). The forest plot is available on Figure S3.

Although initially prespecified, we were not able to pool the results for the association of PH with Apgar score, length of NICU stay, days of parenteral nutrition and need for any feeding tube.

Quality Assessment:

As shown in Table 3, four studies (Hutson 2017[16], Liu 2022[4], Partridge 2014,[5] and Teillet 2022[6]) achieved the maximum score across all NOS domains, indicating low risk of bias. One study (Whitt 2023[8]) met all selection and outcome criteria but did not receive a star for comparability and outcome, due to lack of follow-up and limited data. Overall, the majority of studies demonstrated low risk of bias.

Publication Bias and Risk of Bias Due to Missing Evidence (ROB-ME):

Only five studies were included in our study. As recommended by Cochrane, we did not perform funnel plots to assess publication bias due to the small number of studies[19].

Additionally, we evaluated the risk of bias due to missing evidence using the ROB-ME framework (Supplementary Material). Across all assessed outcomes, there was no indication that eligible studies were missing due to the magnitude or direction of their results. The results matrix showed that all included studies reported outcomes relative to mortality and mechanical ventilation duration, and no evidence of selective non-reporting was identified. Furthermore, our systematic search strategy across multiple databases reduced the likelihood of unidentifiable eligible studies. Overall, the ROB-ME assessment suggested a low risk of bias due to missing evidence for the analyzed outcomes.

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GRADE Assessment:

We performed the GRADE assessment for the certainty of evidence of our primary outcomes (Supplementary Material – Table S1). We judged mortality and duration of MV to be of very low certainty due to the risk of bias and imprecision. We downgraded the certainty of evidence due to the risk of bias because no study in our meta-analysis performed adjusted effect measures, as well as inappropriate measure of outcomes (such as reporting "MV duration" rather than "days of alive and free of MV"). We rated the imprecision as very serious for mortality since our 95% CI OR was > 2.5. We also rated the imprecision as very serious for MV duration since the effect was large and the sample size was limited. As recommended per GRADE, we also considered the risk of publication bias and residual confounding in the "other considerations". Therefore, starting from low risk of certainty since the studies are retrospective, we downgraded the certainty of evidence to very low.

Discussion

This meta-analysis represents, to our knowledge, the most comprehensive evaluation to date of the association between PH and clinical outcomes in neonates with giant omphalocele. We found that PH was significantly associated with increased mortality and with prolonged duration of MV by a mean difference of 21.25 days. These findings underscore that PH is a major predictor of poor prognosis in neonates with giant omphalocele. Additionally, prolonged ventilator dependence may reflect both the severity of PH, and the degree of pulmonary underdevelopment.

We also found that neonates with PH had over six times the odds of death compared to those with no PH, consistent with prior smaller cohort studies identifying PH as a critical determinant of outcome in infants with abdominal wall defects, including omphalocele[1, 3, 7]. While omphalocele is often discussed in the context of associated anomalies, cardiac defects or chromosomal abnormalities[4, 6, 9], our study highlights that PH itself represents a key, independent mortality risk factor. One potential explanation may relate to the high prevalence of pulmonary hypoplasia, as well as pulmonary dysplasia, abnormal pulmonary vascular tone, and remodeling of the vascular smooth muscle[4, 6]. Partridge et al. noted that such vascular abnormalities often persist post-repair, requiring long-term vasodilator therapy, and correlate with abdominal wall defect size and visceral herniation[5]. Dadoun et al. further demonstrated that reduced lung volume on fetal magnetic resonance imaging, rather than defect size alone, predicted mortality and severe respiratory morbidity[10], suggesting that factors such as lung volume, liver herniation, and associated anomalies may play a more critical role in determining prognosis. Similar to congenital diaphragmatic hernia, where pulmonary hypoplasia severity strongly predicts PH and mortality [11, 12], emerging evidence indicates that the degree of pulmonary underdevelopment in omphalocele influences both PH risk and overall prognosis.

Our analysis reported the significantly prolonged duration of MV in infants with giant omphalocele and PH, with a mean difference of 21.25 additional days compared to those with no PH, likely reflecting both the severity of PH and the degree of pulmonary underdevelopment. Prolonged MV not only signals greater disease severity but also contributes to complications such as ventilator-associated pneumonia, barotrauma, bronchopulmonary dysplasia, and increased risk of tracheostomy dependence[2, 23]. Sauthier et al. emphasized that prolonged MV in neonates increases mortality, neurological impairment risk, and delays in achieving oral feeding milestones[23], with downstream consequences including chronic lung disease and neurodevelopmental delays. Furthermore, our analysis did not establish a direct association between prenatal omphalocele characteristics and postnatal outcomes. Notably, Dadoun et al[10] demonstrated that reduced fetal lung volume on fetal magnetic resonance imaging (MRI), with an observed-to-expected total lung volume (O/E TLV) < 50% was strongly associated with mortality, need for intubation, prolonged ventilation and PH, whereas omphalocele size itself showed no prognostic value. These findings suggest that prenatal lung volume assessment, rather than defect size alone, may offer more meaningful prognostic information for perinatal counseling and care planning.

Interestingly, neither gestational age nor birth weight was significantly associated with the presence of PH in our analysis. This suggests that intrinsic pulmonary or vascular abnormalities, rather than prematurity or

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growth restriction, may be more relevant in the pathogenesis of PH in this group. While male sex showed a non-statistically significant trend toward higher PH risk, this observation aligns with broader neonatal literature indicating increased susceptibility to respiratory morbidity in male infants, possibly due to delayed lung maturation and hormonal influences[24].

Limitations:

The small number of studies and overall sample size (n=258) reduces statistical power, increases uncertainty around pooled estimates, and limits the robustness and generalizability of the conclusions. Although this meta-analysis represents, to our knowledge, the most comprehensive synthesis on this topic to date, and incorporates the largest currently available sample, the evidence base remains limited. All included studies were retrospective and non-blinded, and their results were largely unadjusted, making the residual confounding likely. Several clinically relevant variables were incompletely or inconsistently reported across the studies, including use of pulmonary vasodilators (e.g., inhaled nitric oxide, sildenafil), presence and severity of pulmonary hypoplasia, type of surgical repair or use of prosthesis, objective size of the defect and objective prenatal lung metrics such as observed-to-expected lung-to-head ratio (O/E LHR). Importantly, pulmonary hypoplasia and/or associated genetic anomalies were heterogeneously reported or not detailed in several manuscripts. This can be a confounder in the association between PG and mortality reported in our study. Hemodynamically significant congenital heart disease was excluded to reduce confounding, as structural cardiac defects independently influence cardiopulmonary hemodynamics and mortality risks in infants with omphalocele.

Importantly, secondary outcomes, such as feeding milestones, enteral and parenteral nutrition, total length of NICU stay, and post-discharge respiratory or nutritional support were either absent or inconsistently reported, limiting comprehensive risk stratification. Because fewer than three studies provided extractable data for each of these outcomes, they could not be quantitatively synthesized according to conventional meta-analytic standards. We attempted to obtain additional data by contacting corresponding authors; however, the requested information was unavailable or no responses were received. The inability to include these variables limited comprehensive risk stratification and precluded assessment of important short- and long-term clinical outcomes, which may influence the interpretation and generalizability of our findings.

The lack of standardized definitions and echocardiographic protocols for diagnosing PH in neonates with omphalocele (e.g., use of estimated right ventricular pressure via tricuspid regurgitation velocity, septal flattening, or qualitative assessment) may have introduced heterogeneity into our dataset and limited validity of the results. In addition, institutional variation in the management of omphalocele, including surgical strategy and timing, ventilatory strategies and perioperative care, represents another potential source of heterogeneity across studies. Although statistical heterogeneity for mortality and mechanical ventilation duration was low (as reflected by I^2 values and forest plot inspection, it is important to recognize that I^2 does not capture clinical and methodological heterogeneity. Since only non-randomized studies were available and their results were not adjusted, residual confounding is likely to be present. We included studies spanning a wide range of years to increase the sample size; however, temporal changes in pediatric care and institutional practice variation may have influenced outcomes.

Conclusion:

In our study, newborns and infants with giant omphalocele and PH have higher odds of mortality, and longer duration of MV compared to those with no PH. Moreover, future prospective multicenter studies that can attain greater sample sizes are needed to better characterize this association and optimize management strategies for this vulnerable population.

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Statement of Ethics:

A Statement of Ethics is not applicable because this study is based exclusively on published literature.

Conflicts of Interest Statement:

The authors have no conflicts of interest relevant to this article to disclose.

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There was no funding for this study.

Author Contributions:

Dr Larissa Maria Isaac Maximo conceptualized and designed the study, designed the data collection instruments, collected and extracted the data, assisted with the analysis, drafted the initial manuscript, and critically reviewed and revised the manuscript.

Dr Gabriel de Assis de Carlos assisted with collection and extraction of the data, analyzed the data, assisted in drafting the initial manuscript, and critically reviewed and revised the manuscript.

Dr Eduardo Messias Hirano Padrao analyzed the data and performed the statistical analysis, assisted in drafting the initial manuscript, and critically reviewed and revised the manuscript.

Leticia Rocha Campos and Simran Gohil assisted with data collection and analysis and critically reviewed and revised the manuscript.

Luis Gustavo Albuquerque Mello de Oliveira assisted with screening and data collection instruments and critically reviewed and revised the manuscript.

Gisella Chiara Carpi assisted with collection and extraction of the data and critically reviewed and revised the manuscript.

Dr Asaad Beshish and Dr Richard Urbano Garcia Soriano provided mentorship throughout the study and critically reviewed and revised the manuscript in multiple stages of the development and writing.

All authors approved the final manuscript as submitted and agree to be accountable for all aspects of the work.

Data Availability Statement:

All data generated or analyzed during this study are included in this article and its supplementary material files. Further enquiries can be directed to the corresponding author.

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Fig. 1: PRISMA flow of study selection

Fig. 2: Pooled results for mortality in PH versus non-PH

Fig. 3: Pooled results for mean difference in MV days in PH versus non-PH

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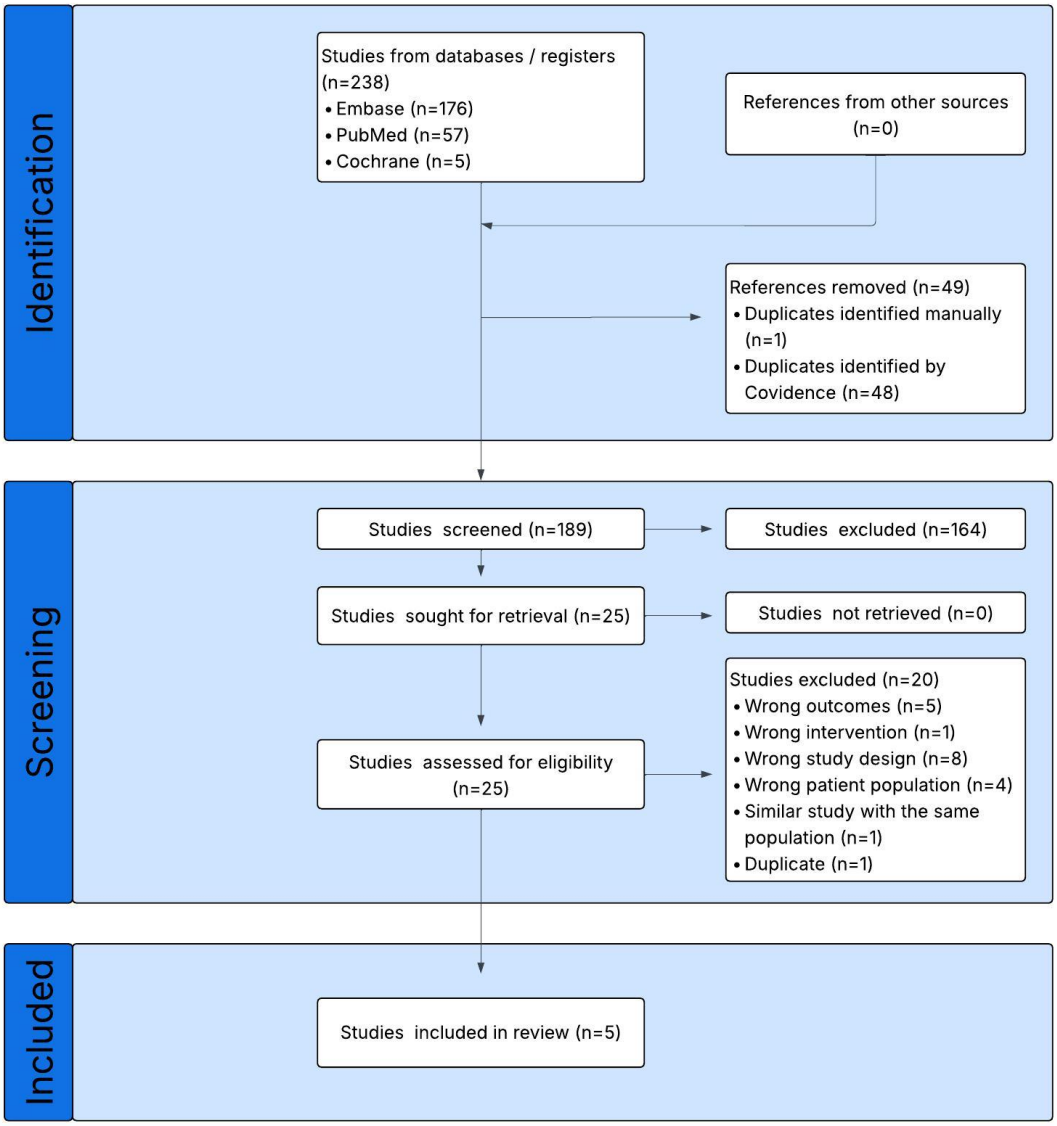

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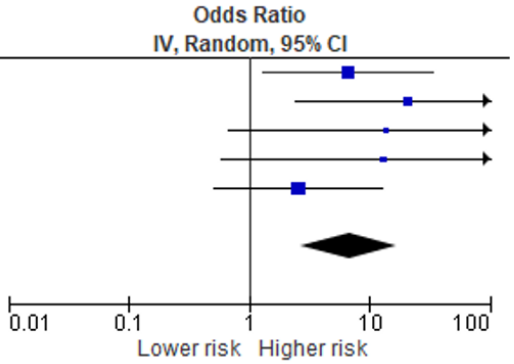
Study or Subgroup	PH Events	PH Total	non-PH Events	non-PH Total	Weight	Odds Ratio IV, Random, 95% CI
Hutson 2017	12	32	2	24	31.9%	6.60 [1.31, 33.18]
Liu 2022	8	24	1	43	17.9%	21.00 [2.43, 181.57]
Partridge 2014	3	20	0	34	9.1%	13.80 [0.67, 282.36]
Teillet 2022	2	15	0	35	8.7%	13.15 [0.59, 291.97]
Whitt 2023	11	21	3	10	32.5%	2.57 [0.52, 12.72]
Total (95% CI)		**112**		**146**	**100.0%**	**6.78 [2.72, 16.88]**
Total events	36		6			

Heterogeneity: Tau² = 0.00; Chi² = 2.86, df = 4 (P = 0.58); I² = 0%

Test for overall effect: Z = 4.11 (P < 0.0001)



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Study or Subgroup	PH Mean	PH SD	PH Total	non-PH Mean	non-PH SD	non-PH Total	Weight	Mean Difference IV, Random, 95% CI
Hutson 2017	81.83	196.29	15	15.5	35.22	27	0.6%	66.33 [-33.89, 166.55]
Liu 2022	19.33	29.16	24	3.33	3.83	43	47.2%	16.00 [4.28, 27.72]
Partridge 2014	45.3	28.9	20	16.5	14.9	34	35.0%	28.80 [15.18, 42.42]
Teillet 2022	23.67	45.8	15	2.67	6.18	35	12.0%	21.00 [-2.27, 44.27]
Whitt 2023	28.81	74.14	21	15.8	25.78	10	5.1%	13.01 [-22.50, 48.52]
Total (95% CI)			**95**			**149**	**100.0%**	**21.25 [13.19, 29.31]**

Heterogeneity: Tau² = 0.00; Chi² = 2.94, df = 4 (P = 0.57); I² = 0%
Test for overall effect: Z = 5.17 (P < 0.00001)

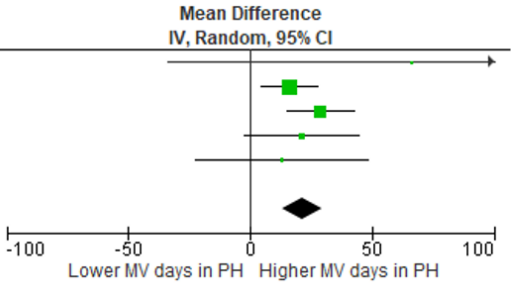

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Table 1: Characteristics of the studies included in the analysis

Author Year	Study Period	Study Design	Inclusion Criteria	Exclusion Criteria	Country	PH diagnostic criteria	Total Participants	PH Group (n)	Non-PH Group (n)
Hutson 2017 [16]	1994–2011	Retrospective Cohort	Infants with omphalocele admitted to the NICU	Congenital heart disease and no echocardiogram for review	USA	Flattening of the interventricular septum and/or tricuspid regurgitant jet with estimated right ventricular pressure >40 mm Hg	56	32	24
Liu 2022 [4]	2015–2020	Retrospective Cohort	Infants with omphalocele admitted to the NICU	(1) small omphalocele (2) ruptured omphalocele pre- admission (3) congenital heart disease except for atrial septal defect, patent foramen ovale, patent ductus arteriosus, or small ventricular septal defect (4) patients with incomplete clinical data	China	Tricuspid regurgitant jet with an estimated systolic pulmonary artery pressure > ⅔ systemic SBP and/or right-to-left or bidirectional flow at ductal and/or atrial level	67	24	43
Partridge 2014 [5]	2004–2012	Retrospective Cohort	Patients with giant omphalocele enrolled in the interdisciplinary Pulmonary Hypoplasia Program	Patients with abdominal wall defects not meeting the definition of giant omphalocele, and patients managed by palliative care at birth	USA	RVSP ≥ 27 mmHg or qualitative features including RV dilation, enlargement or hypertrophy; and septal flattening	54	20	34
Teillet 2022 [6]	1996–2021	Retrospective Cohort	All cases of omphaloceles that underwent pre- and post-natal care	Associated malformations of genetic syndromes, patients with insufficient data, patients treated with primary closure of the septal defect, patients with documented palliative care plan	France	Right-to-left or bidirectional flow in DA; Flattening or paradoxical septum if DA is not present; Mean blood flow velocity < 0.25 m/s in pulmonary arteries; Pulmonary or tricuspid regurgitation gradient > 50% of systemic SBP	50	15	35
Whitt 2023 [8]	1996–2016	Retrospective Cohort	Patients with giant omphalocele	–	USA	PH was not defined in the article	59 (only 31 were included in our meta-analysis	21	10

Abbreviations: DA, ductus arteriosus; NICU: neonatal intensive care unit; PH: pulmonary hypertension; USA, United States of America; RV, right ventricular; RVSP, right ventricular systolic pressure; SBP, systolic blood pressure

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Table 2: Characteristics of the study population among studies included in the analysis

Study Year	Male Sex PH n (%)	Male sex Non-PH n (%)	Mortality PH n (%)	Mortality non-PH n (%)	Birth weight PH (g) [1]	Birth weight non-PH (g) [1]	Gestational Age PH (weeks) [1]	Gestational Age non-PH (weeks) [1]	Days on MV PH [1]	Days on MV Non-PH [1]
Hutson 2017 [16]	32 (15)	24 (12)	12 (37.5)	2 (8.3)	2,660 ±914	2,828±877	35.6±3.4	36.3±3.4	5.5 (0–240)	1.5 (0–45)
Liu 2022 [4]	24 (16)	43 (20)	8 (33.3)	1 (2.3)	2834 ± 690[1]	2693 ± 581	38 (36.3–38.3)	38 (36.1–38.7)	11 (5–42)	3 (1–6)
Partridge 2014 [5]	20 (14)	34 (15)	3 (15)	0 (0)	2557.2 ± 700.8	2555.2 ± 660.5	35.4 ± 2.8	35.9 ± 2.8	45.3 ± 28.9	16.5 ± 14.9
Teillet 2022 [6]	15 (5)	35 (17)	2 (13.3)	0 (0)	2586 ± 792	2838 ± 1000	36 ± 3	37 ± 2	15 (0-56)	0 (0-8)
Whitt 2023 [8]	21(10)	10 (?)	11 (5.2)	3 (30)	2721.5 ± 703.6	2575.1 ± 631.9	36.6 ± 2.2	34.9 ± 2.9	4 (0-13)	3 (0.25-13.8)

Abbreviations: g: grams; MV: mechanical ventilation; PH: pulmonary hypertension
[1] Results depicted in median (interquartile range/IQR), or mean ± SD

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Table 3: Risk of bias assessed by Newcastle-Ottawa Scale

Study (year)	Selection				Comparability	Outcomes		
	1	2	3	4		1	2	3
Hutson 2017 [16]	♦	♦	♦	♦	♦♦	♦	♦	♦
Liu 2022 [4]	♦	♦	♦	♦	♦♦	♦	♦	♦
Partridge 2014 [5]	♦	♦	♦	♦	♦♦	♦	♦	♦
Teillet 2022 [6]	♦	♦	♦	♦	♦♦	♦	♦	♦
Whitt 2023 [8]	♦	♦	♦	♦	–	♦	♦	–

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